

Doc-Researcher: A Unified System for Multimodal Document Parsing and Deep Research

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Abstract

Deep Research systems have revolutionized how LLMs solve complex questions through iterative reasoning and evidence gathering. However, current systems remain fundamentally constrained to textual web data, overlooking the vast knowledge embedded in multimodal documents: scientific papers, technical reports, and financial documents where critical information exists in figures, tables, charts, and equations. Processing such documents demands sophisticated parsing to preserve visual semantics, intelligent chunking to maintain structural coherence, and adaptive retrieval across modalities, which are capabilities absent in existing systems. In response, we present **Doc-Researcher**, a unified system that bridges this gap through three integrated components: (i) **deep multimodal parsing** that preserves layout structure and visual semantics while creating multi-granular representations from chunk to document level, (ii) **systematic retrieval architecture** supporting text-only, vision-only, and hybrid paradigms with dynamic granularity selection, and (iii) **iterative multi-agent workflows** that decompose complex queries, progressively accumulate evidence, and synthesize comprehensive answers across documents and modalities. To enable rigorous evaluation, we introduce **M4DocBench**, the first benchmark for Multi-modal, Multi-hop, Multi-document, and Multi-turn

deep research. Featuring 158 expert-annotated questions with complete evidence chains across 304 documents, M4DocBench tests capabilities that existing benchmarks cannot assess. Experiments demonstrate that Doc-Researcher achieves 50.6% accuracy, 3.4× better than state-of-the-art baselines, validating that effective document research requires not just better retrieval, but fundamentally deep parsing that preserve multimodal integrity and support iterative research. Our work establishes a new paradigm for conducting deep research on multimodal document collections.

CCS Concepts

• Information systems → Novelty in information retrieval.

Keywords

Multimodal Document Parsing, Deep Research, Agentic RAG

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1 Introduction

Recently, agentic retrieval-augmented generation (RAG) systems enable LLMs to retrieve, organize, and present information [31, 43]. Agentic RAG enhances traditional RAG in retrieval planning, adaptive and iterative search and refinement, and context validation [18, 44]. These capabilities underpin recent Deep Research systems [13], which support complex reasoning [14, 17–19] and research workflows [9, 20]. However, current deep research systems [38] remain

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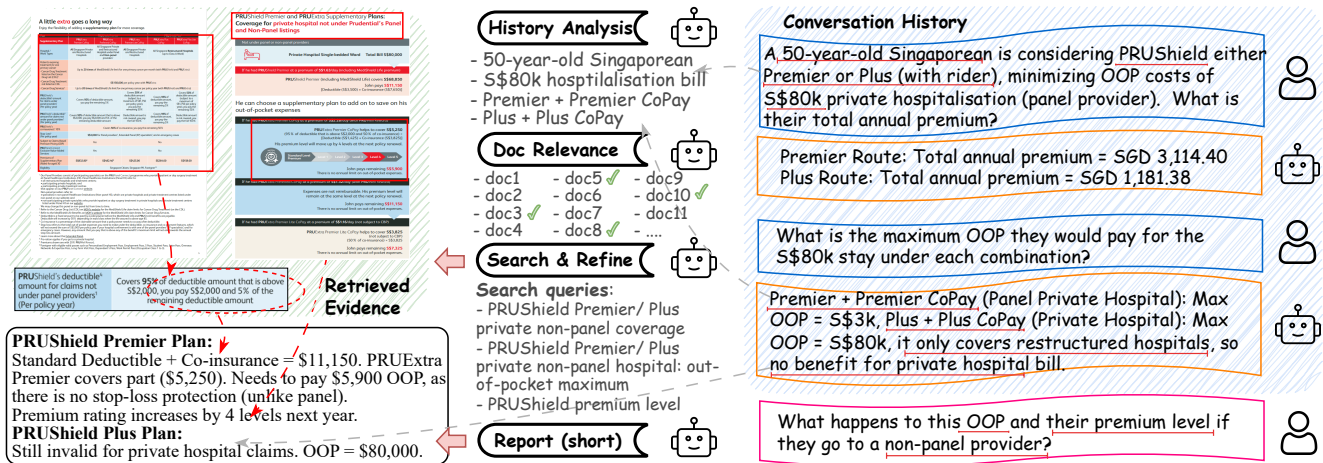


Figure 1: An typical use case of multimodal doc deep research. The user asks a multi-hop question in the context of multi-turn conversations, where ground-truth evidence spans across multiple documents and modalities. The demo conversations, evidence, and answer are shortened for simplicity. Refer to more examples and evidence-chain annotations in Figure 4.

constrained on *textual web data*. They overlook the reality that professional documents are inherently multimodal, seamlessly integrating text with images, tables, equations, and charts [28, 34, 45]. Moreover, these systems lack the capability to process locally stored documents, forcing users to rely solely on public web content rather than local document repositories.

To facilitate multimodal document understanding, recent agentic RAG systems (e.g., MDocAgent [12] and M3DocRAG [3]) have emerged. However, these systems exhibit three critical limitations: (1) **Inadequate multimodal parsing.** Current approaches rely on simplistic parsing strategies: either converting documents into OCR-text [42] or treating them as raw page screenshots [10, 12]. Both methods fail to preserve modality-specific characteristics and rich visual semantics inherent in charts, tables, figures, equations, and complex layouts. (2) **Limited retrieval strategies.** Existing systems employ retrieval mechanisms based solely on text chunks or page screenshots, hindering precise multimodal evidence localization. Also, different research tasks demand varying granularities: document-level summaries [7] for comparative analysis, fine-grained chunks [8] for specific evidence extraction, or visual elements for multimodal understanding. Yet current systems lack mechanisms to dynamically select optimal retrieval strategies across modalities and granularity levels (e.g., full-text, summary, page, chunk) based on query characteristics. (3) **Absence of deep research capabilities.** Current systems are confined to one-shot search rather than multi-step research workflows. Furthermore, no evaluation framework comprehensively assesses realistic multimodal document research, which relies large-scale document collections with annotated evidence chains, multi-hop reasoning, and multi-turn interactions (Figure 1). In contrast, existing benchmarks focus on single-document scenarios [5, 23] (Table 1).

To address these challenges, we introduce **DOC-RESEARCHER**, a unified system that integrates sophisticated multimodal document parsing with multi-agent deep research capabilities. Our approach makes three key contributions: (1) **Deep multimodal parsing**

(§3.1). We develop a comprehensive parsing framework that employs MinerU [40] for layout-aware document analysis, followed by intelligent chunking strategies and multimodal content processing. This framework creates multi-granular document representations that preserve both structural and semantic information, enabling efficient multimodal retrieval. (2) **Systematic retrieval architecture** (§3.2). We conduct extensive experiments on multimodal retrieval paradigms, evaluating 5 text and 5 vision retrievers alongside various reranking and query extension strategies. Our analysis reveals the effectiveness-efficiency tradeoffs among text-only, vision-only, and hybrid approaches. (3) **Deep research workflows** (§3.3). Doc-RESEARCHER supports iterative, multi-step research workflows on local document collections through intelligent planning capabilities. The framework adaptively selects optimal retrieval granularities based on query characteristics and content types, enabling complex multi-hop reasoning across documents.

Furthermore, we introduce **M4DocBENCH** (§4), a comprehensive benchmark for evaluating multimodal, multi-hop, multi-doc, and multi-turn deep research. It provides expert-annotated QA pairs with complete evidence chains, enabling rigorous assessment of document understanding in realistic research scenarios. Experimental results demonstrate that Doc-RESEARCHER significantly outperforms existing approaches, establishing a new paradigm for multimodal document retrieval and deep research. In summary, our contributions are threefold: (1) **Deep multimodal parsing and retrieval framework:** We develop modular, plug-and-play components for multimodal document parsing and retrieval that can enhance any deep research system, featuring layout-aware analysis and multi-granular representation strategies. (2) **Doc-RESEARCHER system:** We present the first unified deep research workflows, leveraging adaptive and granular retrieval, and iterative evidence refinement for multimodal documents understanding. (3) **M4DocBENCH benchmark:** We establish the first comprehensive benchmark for multimodal deep research, featuring large-scale document collections with annotated evidence chains for multi-hop, multi-document, and multi-turn evaluation.

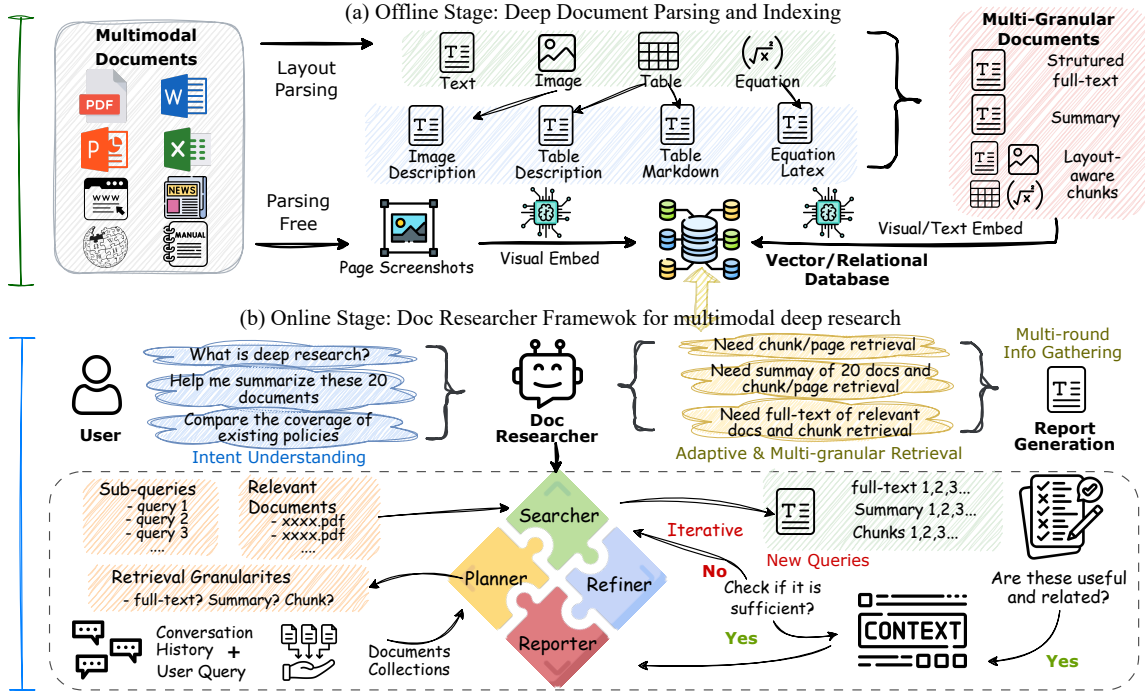


Figure 2: DOC-RESEARCHER Architecture: (a) multimodal deep parsing and indexing, and (b) multimodal deep research.

2 Related Work

Document Parsing Strategies. Document parsing approaches fall into three categories, each with distinct tradeoffs. **Shallow parsing** relies on OCR to extract text while discarding spatial and visual information [32]. **Deep parsing** systems like MinerU [40] preserve layout structure through bounding boxes, transcribe equations to LaTeX [39], and convert tables to structured formats. Recent VLM-based parsers [21, 27] further enhance extraction quality. **Parsing-free** approaches [10, 22] bypass parsing entirely, processing documents as page screenshots. While avoiding parsing overhead, they sacrifice fine-grained localization and struggle with dense text. Our deep parsing framework synthesizes these approaches' strengths: preserving visual semantics like parsing-free methods while maintaining structural precision of deep parsing.

Visual Document Understanding. DocVQA benchmarks evolve from single-page tasks [25, 26] to multi-hop reasoning across pages [15, 37] and recently to long documents [23, 47]. However, these benchmarks evaluate isolated capabilities: either VLM long-context processing (1-20 images) or single-document QA (50+ pages). Crucially, none assess multi-document synthesis, multi-turn interaction, or evidence chain annotation needed for research-level tasks. Hence, we propose M4DocBENCH (Section 4) to address these gaps with comprehensive deep research evaluation.

Document RAG Systems. Due to limited context windows, DocRAG systems have merged that decompose understanding into retrieval and reasoning stages. Current systems [3, 6, 12] employ various strategies: M3DocRAG uses vision-only page retrieval, MDocAgent combines text chunks with screenshots, and VDocRAG [33] incorporates figure extraction. However, all perform single-round retrieval with fixed granularities, lacking the iterative

refinement and adaptive strategies essential for complex research. DOC-RESEARCHER introduces the first deep research framework for multimodal documents, enabling multi-step investigation through dynamic granularity selection and progressive evidence synthesis.

3 Method: DOC-RESEARCHER

DOC-RESEARCHER (in Figure 2) consists of two main modules: deep parsing and deep research. The deep parsing pipeline transforms multimodal documents into structured, multi-granular, and searchable units while preserving multimodal semantics. The deep research adopts a multi-agent collaborative framework to solve complex questions with iterative search and refinement.

3.1 Deep Multimodal Document Parsing

Multi-modal Doc Parsing. Given a collection of multimodal documents $\mathcal{D}$, each document $d_i \in \mathcal{D}$ undergoes layout parsing via MinerU [40] to extract structural elements $E_i = \{e_{i,j,k}\}$, where indices denote document i , page j , and reading sequence k . Each element is pair with its layout type (text, table, figure, equation), and bounding box coordinates (Figure 5a). For computational efficiency, we perform one-time textual conversion of visual elements. Tables and figures are processed through Qwen2.5-VL [1] to generate both coarse-grained summaries for contextual understanding and fine-grained descriptions for detailed content capture. Equations are converted to LaTeX format using UniMERNet [39]. This preprocessing strategy enables efficient retrieval and reasoning operations while avoiding repeated multimodal token processing.

Multi-granular Doc Chunking. While individual layout elements E_i preserve structural integrity, they often lack sufficient context. We address this through layout-aware chunking that merges

text elements within section boundaries, subject to maximum length constraints (Figure 5b). This strategy maintains semantic coherence while preserving full traceability via page IDs and bounding box coordinates for precise localization and citation. We construct 4 different granularity levels $G = \{\text{chunk, page, full, summary}\}$: (1) chunks created through layout-aware merging, (2) pages by either combining all elements $e_{i,j}$ within page j , or raw page screenshots, (3) full-text containing all elements within document d_i , and (4) document summary by summarizing full-text using LLMs.

3.2 Multimodal Retrieval Architecture

Vision-only Retrieval. Vision-only retrieval operates at page granularity using raw screenshots can eliminate parsing overhead of OCR or layout analysis. However, it faces challenges in representing complex pages via dense vectors. For high-resolution pages, multi-vector vision retrievers can produce thousands of vectors, where each small patch (typically 14x14 pixels) correspond to a vector. Additionally, vision retrievers rely on parameter-heavy VLMs (often >3B parameters), increasing computational costs.

Text-only Retrieval. Text-only retrieval supports both page and chunk granularities, encoding text chunks, and OCR-extracted or VLM-described text of visual elements. This approach leverages lightweight models (often <1B parameters) that excel at encoding text-intensive documents. However, the OCR or VLM preprocessing incurs costs and may lose visual semantics.

Hybrid Retrieval. Hybrid retrieval combines strengths of both paradigms by directly encoding visual chunks without intermediate textual conversion, preserving rich visual information while maintaining text understanding capabilities. This approach eliminates information loss from OCR or VLM description but requires more computational resources during inference.

3.3 Multimodal Deep Research

Strategic Document Filtering and Adaptive Processing. Given a query q_i with dialog history h_i , the Planner addresses computational efficiency in large-scale document analysis via intelligent document selection and determining retrieval granularity. The agent analyzes (q_i, h_i) to produce three outputs: (i) a filtered document subset $\mathcal{D}' \subseteq \mathcal{D}$ by matching query semantics against document summaries, reducing search space by 60-80% while maintaining high recall; (ii) optimal retrieval granularity $\theta \in G = \{\text{summary, chunk, full, summary}\}$ based on query characteristics; and (iii) refined sub-queries at timestamp $Q_t = \{\tilde{q}_1, \dots, \tilde{q}_n\}$ that decompose complex questions or explore in new search directions. This adaptive strategy leverages the multi-granular representations from our parsing framework (§3.1). For instance, broad contextual queries activate summary mode to access document overviews directly, while technical queries that require specific evidence trigger targeted chunk extraction through iterative refinement.

Iterative Search-Refine Loop. The core research process operates through dynamic collaboration between Searcher and Refiner. For each sub-query $\tilde{q}_k \in Q_t$, the system executes an iterative loop:

$$\mathcal{R}_t = \text{Search}(\tilde{q}_k, \mathcal{D}', \theta), \quad \mathcal{R}_t^* = \text{Refine}(\mathcal{R}_t, \tilde{q}_k) \quad (1)$$

$$\sigma_t = \text{Evaluate}(\mathcal{R}_1^* \cup \dots \cup \mathcal{R}_t^*, q_i) \quad (2)$$

where $\mathcal{R}_t$ represents content retrieved at iteration t using our multimodal retrieval architecture (§3.2), $\mathcal{R}_t^*$ is the refined relevant subset after deduplication and relevance filtering, and σ_t measures information sufficiency. The search stops when either sufficiency threshold is met ($\sigma_t \geq \tau$) or maximum iterations reached ($t = T_{\max}$), balancing thoroughness with computational efficiency. If new search is required, subqueries set is updated concurrently ($Q_t \rightarrow Q_{t+1}$).

Multimodal Report Generation. The Reporter synthesizes accumulated evidence $\mathcal{R}^* = \bigcup_t \mathcal{R}_t^*$ into a comprehensive response $\hat{a} = \text{Report}(q_i, \mathcal{R}^*, C)$, where C contains citation metadata (page IDs and bounding boxes coordinates) from our parsing framework. By analyzing query intent and retrieved information, the Reporter generates interleaved text-image outputs that directly incorporate relevant visual elements (e.g., tables, figures, and charts) alongside textual explanations. This supports the notion that “a single image is worth a thousand words”. Moreover, the multimodal composition enhances both answer quality and verifiability, as users can directly verify claims to precise document locations via embedded multimodal citations. The approach proves particularly effective for complex analytical queries requiring evidence synthesis across diverse multimodal sources.

4 M4DocBENCH Benchmark

4.1 Motivation of M4DocBENCH

A comprehensive evaluation must assess four critical dimensions: (1) **Multi-hop reasoning**: Complex questions requiring sophisticated reasoning chains across evidence. (2) **Multi-modal integration**: Questions demanding information from at least two modalities (text, tables, images). (3) **Multi-doc synthesis**: Questions relying evidence from multiple documents, where each provides unique, essential information. Distractor documents with topically related content increase retrieval difficulty. (4) **Multi-turn interaction**: Dialogical contexts reflecting real-world research sessions, where current queries require information and disambiguation of prior exchanges. Table 1 reveals critical gaps in existing benchmarks. To address these limitations, we introduce **M4DocBENCH**, a benchmark specifically designed for Multi-modal, Multi-hop, Multi-document, and Multi-turn deep research evaluation. M4DocBENCH features expert-annotated question-answer pairs with complete evidence chains, enabling rigorous assessment of document understanding capabilities in realistic research scenarios.

4.2 Domain and Document Collection

We select four distinct domains based on three criteria: (1) **Multi-modal richness**: Each domain features essential visual elements integrated with text; (2) **Reasoning complexity**: Questions require diverse comprehension from technical to numerical analysis; (3) **Practical relevance**: These represent real-world scenarios where users conduct routine research. We collect documents from authoritative sources to ensure quality and representativeness:

- **Research**: We collect 166 papers from arXiv and top-tier conferences, averaging 26.7 pages and 21.8k words in length.
- **Insurance**: We gather 43 brochures and policy documents from major insurance insurers, averaging 11.0 pages and 5.5k words.

Benchmarks	Question of multi-hop modal doc turn				QA	Annotation Type	Domain	lang'	Retrieval Annotation						
									doc	#doc	page	#page	layout	#lay	subquery
ViDoRe [10]	✗	✓	✗	✗	3,810	Existing+Claude3	Multiple	2	✗	1.0	✓	1.0	✗	-	✗
MMLongBench-Doc [23]	✓	✓	✗	✗	1,082	Expert	7 domains	1	✗	1.0	✓	1.2	✗	-	✗
DocBench [47]	✓	✓	✗	✗	1,102	GPT4 + Human	5 domains	1	✗	1.0	✗	-	✗	-	✗
M3SciQA [16]	✗	✓	✗	✗	1,452	GPT-4 + Human	NLP papers	1	✓	2.0	✗	-	✗	-	✗
M3DocVQA [3]	✗	✓	✗	✗	2,441	Human	Wikipedia	1	✗	1.0	✓	1.4	✗	-	✗
ViDoSeek [42]	✓	✓	✗	✗	3,162	Human	Slide	1	✗	1.0	✓	1.0	✗	-	✗
MMDocIR [5]	✓	✓	✗	✗	1,658	Expert	10 domains	1	✗	1.0	✓	1.5	✓	1.8	✗
ViDoRe-v2 [24]	✓	✓	✓	✗	271	GPT-4o + Human	ESG/Bio/Eco	4	✗	1.0	✓	5.1	✗	-	✗
MMDocRAG [6]	✓	✓	✗	✗	4,055	GPT-4o + Human	10 domains	1	✗	1.0	✓	1.4	✓	1.7	✗
Double-Bench [30]	✓	✓	✗	✗	5,168	KG+GPT-4o+Human	Multiple	6	✗	1.0	✓	5.9	✗	-	✓
M4DocBENCH	✓	✓	✓	✓	158	Expert	4 domains	2	✓	3.8	✓	7.0	✓	14.8	✓

Table 1: M4DocBENCH compared to other Document RAG/VQA dataset.

- **Education:** We collect 56 program materials (averaging 7.5 pages and 3.9k words) from prestigious universities, including merged program webpages and separate supplementary documents.
- **Finance:** We obtain 38 market research reports (22.2 pages and 11.2k words on average), covering monthly and half-yearly trends.

4.3 Annotation Process

Our annotation process employs PhD/Master-level researchers with extensive research experience to ensure benchmark quality. Beyond basic QA pairs, we annotate six critical dimensions as follows:

- **Hard Negative Documents:** We carefully curate documents that share topical similarity with target questions while lacking essential information. This tests systems' ability to distinguish truly relevant sources from plausible distractors.
- **Retrieval Granularity:** We label optimal retrieval granularity based on query characteristics. For example, single-doc summarization requires *full-text* retrieval, while multi-document synthesis relies *summary* or *chunk* retrieval due to context constraints. This enables evaluation of adaptive granularity selection.
- **Fine-grained Layout:** Following MMDocIR [5], we annotate both relevant pages and bounding boxes within those pages, enabling precise evaluation at element levels.
- **Subquery Decomposition:** We provide ground-truth intermediate queries to reveal decomposition strategies. These annotations enable evaluation of query planning and refinement capabilities.
- **Answer Verification Checklists:** For lengthy responses where objective evaluation proves challenging, we develop structured checklists with clear criteria for assessing answer completeness and accuracy, enabling systematic LLM-based evaluation.
- **Bilingual Support:** All annotators are proficient in English and Chinese, enabling annotation for bilingual capabilities.

To facilitate understanding on M4DocBENCH, we demonstrate one annotation example in Figure 1 and two examples in Figure 4.

4.4 Benchmark Statistics

Table 5 presents statistics of M4DocBENCH, demonstrating its unique evaluation capabilities: **(1) Doc-scale Complexity.** Questions require processing 12.7 documents (306.4 pages) on average, up to 42 documents (836 pages) in challenging cases. Systems must identify 3.8 relevant documents from extensive collections, testing critical document filtering capabilities underexplored in existing benchmarks. **(2) Multimodal and Complex Evidence Requirements.** 112 questions require multiple modalities, with balanced

evidence distribution across text (40.9%), tables (30.2%), and images (28.9%). Fine-grained evidence annotations average 7.0 pages and 14.8 layout elements per question, indicating the complexity of M4DocBENCH. **(3) Multi-turn Depth.** 58 questions occur in multi-turn conversations, capturing iterative research workflows. Questions average 3.9 search clauses and answers range from 59.7 to 272.0 words, testing adaptive query planning and final answer.

5 Experiment

5.1 Evaluation Metrics and Benchmarks

Multimodal Retrieval Evaluation. We evaluate retrieval performance at three granularities: document, page, and layout. For each granularity, we compute recall@k by comparing retrieved passages¹ against gold annotations. Document and page recall are computed using exact ID matching. For chunk-level evaluation, we follow Dong et al. [5] to calculate recall based on bounding box overlap between retrieved and gold-standard layouts. Note that parsing-free methods cannot be evaluated at chunk granularity.

Deep Research Evaluation. We evaluate the complete deep research workflow (§3.3) across three dimensions: **(1) Document Selection:** assessing Planner's ability to filter relevant documents from noisy collections, computing precision, recall, and F₁ scores against gold document sets. **(2) Agentic Retrieval:** After iterative search, we evaluate the accumulated passages using the same metrics in multimodal retrieval, but with variable passage counts reflecting the system's adaptive selection. **(3) Answer Accuracy:** We use LLM-as-judge with annotated checklists. Answers are marked correct only when all checklist items are satisfied.

Evaluation Benchmarks. (1) **M4DocBENCH** (§4) for comprehensive evaluation of both multimodal retrieval and deep research; (2) **MMDocIR** [5] containing 1,658 single-document VQA questions for focused evaluation of page and chunk-level retrieval.

5.2 System Setting and Baselines

Retrieval System. We evaluate 10 retrievers to implement our multimodal retrieval architecture (§3.2): **5 Text retrievers:** BM25 [29], E5 [41], BGE-M3 [2], Qwen3-embedding-0.6B [46], and Qwen3-reranker-0.6B [46]. **5 Vision retrievers:** DSE [22], ColPali [10], ColQwen [10], and Jina-embedding-v4 [11] (multi-vector and dense variants). We configure retrievers in three paradigms: **(1) Text-only:** Text retrievers encode both textual and visual chunks, with

¹Passages denote: chunks for parsed methods, and pages for parsing-free approaches.

	Model	Parameter	Embed type dimension	M4DocBench			MMDocIR				
				doc	page	layout	page	layout			
Recall@k = 10	Text	BM25	-	-	51.5 / 59.3	22.1 / 25.6	15.8 / 17.1	60.0 / 65.0	45.1 / 48.8		
		E5	0.34B	dense	1,024	65.7 / 75.4	29.9 / 31.8	18.0 / 18.0	54.4 / 61.3	38.0 / 44.8	
		BGE-M3	0.56B	dense	1,024	75.4 / 81.4	33.3 / 40.0	19.2 / 20.8	58.0 / 61.9	41.3 / 44.9	
		Qwen3-embedding	0.60B	dense	1,024	73.6 / 78.8	34.6 / 40.6	20.1 / 23.5	58.6 / 62.6	42.6 / 45.6	
		Multi+reranking*	1.5B	hyrbid	-	78.7 / 80.3	39.2 / 40.9	23.1 / 21.1	65.5 / 66.7	48.6 / 49.4	
	Vision	DSE _{wiki-ss}	4.15B	dense	3,072	34.8 / 42.4	11.1 / 11.7	-	68.4 / 68.7	-	
		ColPali	2.92B	multi-vec	$N_{\text{tok}} \times 128$	47.7 / 50.0	18.1 / 17.9	-	75.4 / 73.9	-	
		ColQwen	2.21B	multi-vec	$N_{\text{tok}} \times 128$	49.8 / 55.8	25.5 / 24.2	-	77.4 / 75.4	-	
		Jina-embedding-v4	3.75B	dense	2,048	52.9 / 56.3	23.6 / 22.9	-	74.4 / 71.5	-	
		Jina-embedding-v4	3.75B	multi-vec	$N_{\text{tok}} \times 128$	54.3 / 58.7	25.8 / 26.7	-	80.8 / 79.4	-	
	Hyb'd	Jina-embedding-v4	3.75B	dense	2,048	74.0 / 79.4	39.1 / 40.5	28.7 / 27.9	83.4 / 84.0	66.1 / 65.8	
		Qwen3 + Jina	0.60/3.75B	dense	1,024/2,048	74.0 / 79.6	37.5 / 37.5	28.1 / 26.9	82.9 / 83.3	65.4 / 64.7	
		BGE-M3 + Jina	0.56/3.75B	dense	1,024/2,048	77.1 / 81.8	38.4 / 39.1	28.1 / 27.5	83.7 / 83.9	66.2 / 66.0	
	Recall@k = 15	Text	BM25	-	-	58.1 / 66.3	28.0 / 32.6	20.3 / 22.9	62.4 / 68.0	46.8 / 52.2	
			E5	0.34B	dense	1,024	75.4 / 81.9	35.3 / 41.7	22.1 / 25.2	58.0 / 64.4	41.2 / 48.0
			BGE-M3	0.56B	dense	1,024	83.3 / 87.6	41.2 / 46.8	23.8 / 24.9	61.4 / 64.8	44.4 / 47.9
			Qwen3-embedding	0.60B	dense	1,024	81.1 / 86.8	42.1 / 49.9	25.4 / 27.4	61.4 / 65.3	45.2 / 48.3
			Multi+reranking*	1.5B	hybrid	-	83.6 / 87.2	44.8 / 49.4	27.4 / 27.9	67.9 / 70.1	51.0 / 53.0
Vision		DSE _{wiki-ss}	4.15B	dense	3,072	45.2 / 53.9	18.1 / 18.5	-	78.0 / 78.5	-	
		ColPali	2.92B	multi-vec	$N_{\text{tok}} \times 128$	55.7 / 61.7	24.7 / 24.9	-	82.0 / 80.9	-	
		ColQwen	2.21B	multi-vec	$N_{\text{tok}} \times 128$	60.6 / 69.4	33.8 / 33.3	-	83.1 / 82.9	-	
		Jina-embedding-v4	3.75B	dense	2,048	64.7 / 70.9	32.2 / 33.0	-	80.3 / 79.4	-	
		Jina-embedding-v4	3.75B	multi-vec	$N_{\text{tok}} \times 128$	66.0 / 74.4	35.2 / 38.8	-	85.8 / 84.7	-	
Hyb'd		Jina-embedding-v4	3.75B	dense	2,048	83.6 / 87.6	49.2 / 49.9	35.6 / 35.2	87.9 / 87.9	70.6 / 70.5	
		Qwen3 + Jina	0.60/3.75B	dense	1,024/2,048	83.9 / 86.5	46.0 / 47.3	33.5 / 33.7	87.0 / 87.3	69.7 / 69.3	
		BGE-M3 + Jina	0.56/3.75B	dense	1,024/2,048	84.1 / 87.0	44.9 / 47.1	33.4 / 34.1	88.5 / 88.1	70.9 / 70.4	
Recall@k = 20		Text	BM25	-	-	61.3 / 69.3	30.2 / 36.3	22.1 / 24.8	64.5 / 65.3	48.9 / 49.5	
			E5	0.34B	dense	1,024	79.4 / 85.1	39.4 / 47.5	25.6 / 29.0	60.1 / 66.4	43.3 / 49.7
			BGE-M3	0.56B	dense	1,024	86.6 / 89.9	46.8 / 52.3	27.3 / 28.7	62.8 / 67.0	45.5 / 49.8
			Qwen3-embedding	0.60B	dense	1,024	84.9 / 90.6	47.2 / 56.0	30.5 / 30.4	63.5 / 66.9	47.0 / 50.2
			Multi+reranking*	1.5B	hybrid	-	87.8 / 90.3	50.6 / 56.3	31.9 / 32.3	70.1 / 71.7	53.5 / 54.9
	Vision	DSE _{wiki-ss}	4.15B	dense	3,072	52.5 / 59.5	23.3 / 23.5	-	83.1 / 83.8	-	
		ColPali	2.92B	multi-vec	$N_{\text{tok}} \times 128$	61.5 / 67.8	29.5 / 30.0	-	86.0 / 85.6	-	
		ColQwen	2.21B	multi-vec	$N_{\text{tok}} \times 128$	67.8 / 77.0	40.0 / 40.9	-	86.8 / 86.9	-	
		Jina-embedding-v4	3.75B	dense	2,048	73.0 / 78.2	39.7 / 41.3	-	84.6 / 84.0	-	
		Jina-embedding-v4	3.75B	multi-vec	$N_{\text{tok}} \times 128$	70.9 / 80.5	42.1 / 46.3	-	88.9 / 88.4	-	
	Hyb'd	Jina-embedding-v4	3.75B	dense	2,048	88.6 / 89.9	55.5 / 56.5	40.9 / 40.6	90.0 / 90.7	73.4 / 73.6	
		Qwen3 + Jina	0.60/3.75B	dense	1,024/2,048	87.2 / 90.1	53.1 / 54.5	39.4 / 39.7	89.8 / 89.7	72.2 / 72.5	
		BGE-M3 + Jina	0.56/3.75B	dense	1,024/2,048	88.2 / 89.9	50.9 / 53.8	38.4 / 39.5	90.7 / 90.7	73.7 / 73.9	

Method	Backbone	Retriever	Parsing Level	Accuracy								Doc Selection			Retrieval Recall			
				All	en	zh	ins'	res'	edu'	fin'	Rec	Prec	F ₁	Doc	Page	Lay'	#psg	
Direct	Qwen3-32B	-	-	7.0	3.8	10.1	8.3	10.3	0.0	5.6	-	-	-	-	-	-	-	
Direct	Qwen3-235B	-	-	5.1	5.1	5.1	4.2	7.4	3.3	2.8	-	-	-	-	-	-	-	
Direct	DeepSeek-R1	-	-	10.1	13.9	6.3	20.8	11.8	10.0	0.0	-	-	-	-	-	-	-	
Long-context	Qwen3-32B	-	Shallow	13.3	19.0	7.6	25.0	20.6	0.0	2.8	-	-	-	-	-	-	-	
Long-context	Qwen3-235B	-	Shallow	8.9	5.1	12.7	12.5	14.7	3.3	0.0	-	-	-	-	-	-	-	
Long-context	DeepSeek-R1	-	Shallow	29.1	29.1	29.1	20.8	47.1	20.0	8.3	-	-	-	-	-	-	-	
Long-context	Qwen3-32B	-	Deep	13.3	12.7	13.9	20.8	22.1	3.3	0.0	-	-	-	-	-	-	-	
Long-context	Qwen3-235B	-	Deep	11.4	7.6	15.2	16.7	19.1	3.3	0.0	-	-	-	-	-	-	-	
Long-context	DeepSeek-R1	-	Deep	31.7	34.2	29.1	33.3	45.6	30.0	5.6	-	-	-	-	-	-	-	
MDocAgent	InternVL3.5-38B	Hybrid (Multi) [▽]	Shallow	15.8	19.0	12.7	20.8	23.5	6.7	5.6	-	-	-	74.22	37.54	-	10	
M3DocRAG	InternVL3.5-38B	Vision (ColPali)	Free	7.0	10.1	3.8	4.2	2.9	26.7	0.0	-	-	-	69.58	34.35	-	10	
Colqwen-gen	InternVL3.5-38B	Vision (ColQwen)	Free	5.7	6.3	5.1	0.0	8.8	10.0	0.0	-	-	-	74.38	46.29	-	10	
DOC-RESEARCHER	Qwen3-32B [★]	Vision (Jina)	Free	36.7	34.2	39.2	20.8	52.9	40.0	13.9	-	-	-	88.9	61.3	-	17.9	
DOC-RESEARCHER	Qwen3-235B [★]	Vision (Jina)	Free	36.7	36.7	36.7	29.2	50.0	40.0	13.9	-	-	-	<u>89.0</u>	61.0	-	17.9	
DOC-RESEARCHER	DeepSeek-R1 [★]	Vision (Jina)	Free	43.7	45.6	41.7	37.5	61.8	43.3	13.9	-	-	-	90.6	63.6	-	17.4	
DOC-RESEARCHER	Qwen3-32B	Text (Multi) [★]	Shallow	34.2	37.9	30.4	12.5	55.9	36.7	5.6	-	-	-	88.2	59.8	-	14.7	
DOC-RESEARCHER	Qwen3-235B	Text (Multi) [★]	Shallow	36.1	35.4	36.7	12.5	54.4	46.7	8.3	-	-	-	88.2	60.5	-	14.7	
DOC-RESEARCHER	DeepSeek-R1	Text (Multi) [★]	Shallow	39.2	34.2	44.3	20.8	57.4	50.0	8.3	-	-	-	87.5	61.1	-	14.5	
DOC-RESEARCHER	Qwen3-32B	Text (Multi) [★]	Deep	39.2	41.8	36.7	20.8	64.7	26.7	13.9	84.2	75.9	76.0	80.8	59.7	35.1	14.2	
DOC-RESEARCHER	Qwen3-235B	Text (Multi) [★]	Deep	45.6	43.0	48.1	<u>41.7</u>	63.2	50.0	11.1	88.0	77.0	78.4	82.9	60.6	37.7	14.8	
DOC-RESEARCHER	DeepSeek-R1	Text (Multi) [★]	Deep	45.6	44.3	46.8	33.3	67.7	46.7	11.1	89.0	79.1	79.8	82.6	59.2	38.7	12.9	
DOC-RESEARCHER	Qwen3-32B	Hybrid (Jina)	Deep	42.4	43.0	41.8	20.8	70.6	33.3	11.1	87.0	78.1	78.8	83.2	59.4	<u>43.1</u>	17.2	
- w/o planner	Qwen3-32B	Hybrid (Jina)	Deep	36.7	39.2	35.4	29.2	50.0	40.0	11.1	-	-	-	85.7	58.1	40.5	14.5	
DOC-RESEARCHER	Qwen3-235B	Hybrid (Jina)	Deep	<u>47.5</u>	<u>46.8</u>	48.1	<u>41.7</u>	64.7	<u>53.3</u>	13.9	86.7	79.1	<u>79.4</u>	82.2	61.2	45.7	15.8	
- w/o planner	Qwen3-235B	Hybrid (Jina)	Deep	39.2	35.4	43.0	26.7	57.4	43.3	13.9	-	-	-	88.0	58.2	41.2	13.9	
DOC-RESEARCHER	DeepSeek-R1	Hybrid (Jina)	Deep	50.6	53.2	48.1	45.8	70.6	56.7	11.1	89.5	77.4	78.6	82.9	58.0	41.0	14.1	
- w/o planner	DeepSeek-R1	Hybrid (Jina)	Deep	42.9	44.0	41.8	31.2	61.7	43.3	8.3	-	-	-	84.2	55.3	38.8	13.6	

Table 3: Main results on M4DocBENCH. The best score is in **boldface and colored and second best is underlined. [▽] means BGE-M3 that for text retrieval and ColQwen for page retrieval. * refers to using Qwen3-embed, BGE-M3, and E5 for text retrieval. * indicates using InternVL3.5-38B to extract relevant information from raw screenshots, as LLM backbones cannot read images.**

Granularity-Performance Tradeoff. Performance degrades significantly with finer granularity: document-level recall exceeds 70% for hybrid methods, but page- and layout-level recall drop by 30-40%. This indicates that coarse-grained tasks are relatively easy to satisfy, whereas fine-grained retrieval task requires more adaptive and multi-granular retrieval strategies.

5.4 Main Results: Multimodal Deep Research

Table 3 validates our core contributions via systematic comparisons.

Deep Research can largely improve multimodal understanding. Direct answering achieves only 5-10%; long-context processing ranges within 9-31% despite document content of 96k tokens is given. Results show that neither parametric knowledge nor brute-force context extension can handle research-level complexity. In comparison, DOC-RESEARCHER with multi-agent framework (with strategic planning, iterative search-refine loops, and progressive evidence accumulation) achieves score of 50.6%. The 20-40% gap over long-context proves that deep research needs fundamentally multi-agent workflows, not just more context.

Deep Parsing Outperforms Simplistic Parsing. Relying on shallow parsing, DOC-RESEARCHER achieves 34-39% accuracy, which is worse than using parsing-free screenshot processing (36-43%). This confirms that simplistic OCR destroys visual semantics. DOC-RESEARCHER equipped with deep parsing and hybrid retrieval, which preserves layout and multimodal characteristics, reaches 42-50%. The 10% absolute gain validates that modality-specific preservation via deep parsing is essential for document understanding.

Iterative Workflows Enable Deep Research. Baseline RAG systems fail due to single-round and rigid retrieval: MDocAgent

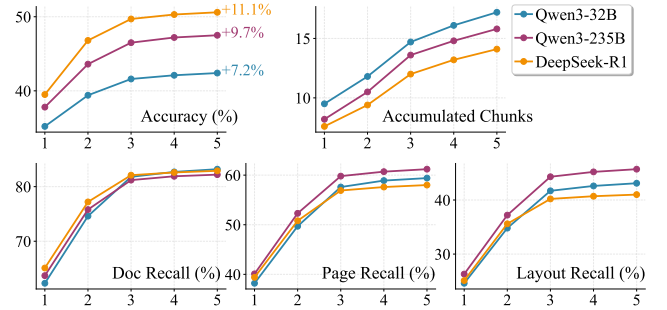


Figure 3: Performance with increasing search depth.

(15.8%) cannot adaptively select strategies despite using dual modalities; M3DocRAG (7.0%) and ColQwen-gen (5.7%) are limited to page-level vision retrieval. In contrast, DOC-RESEARCHER achieves 50.6% which is 3.4× better than MDocAgent, as it allows for the dynamic decomposition and refinement of queries over multiple steps, enabling the multi-hop reasoning and evidence synthesis.

5.5 Ablation Study

We ablate components of DOC-RESEARCHER, and validate different architectural design as follows:

Hybrid Retrieval Necessity. Within our deep parsing framework, hybrid retrieval (42.4-50.6%) outperforms text-only (39.2-45.6%) by 3-5%. This shows that even with high-quality textual transcriptions in both coarse and fine-grained level, some visual semantics is lost. In comparison, visual encoders that directly encode using chunk image, can capture irreplaceable semantics. This validates our hybrid retrieval design over single-modality approaches.

Parsing Setting			Offline Stage: Doc Parsing Latency and Cost of M4DocBENCH corpus				Online Stage: DOC-RESEARCHER Latency per M4DocBENCH question				
Level	Modality	Type	Parsing	Indexing	Captioning	Storage (GB)	Planner	Searcher	Refiner	Reporter	Sum
Parsing-Free	Vision	Dense	-	08:56	-	0.15	-	2.11	07:32	01:20	08:54
	Vision	Multi-vec	-	39:14	-	4.33	-	22.70	07:48	01:24	09:35
Shallow-parsing	Text	Dense	18:47	09:50	-	0.37	-	3.99	01:20	00:45	02:09
Deep-parsing	Text	Dense	01:21:21	12:47	01:12:32	0.34	41.22	4.14	01:37	00:56	03:18
	Hybrid	Dense	01:21:21	14:40	01:12:32	0.43	46.05	0.88	01:40	00:49	03:16

Table 4: Efficiency analysis on different parsing setting and their effect on DOC-RESEARCHER systems. The latency is measured by HH:MM:SS and storage is in GB.

Adaptive Planning Impact. Removing the Planner causes the accuracy drop of 6-8%. Without it, the system defaults to uniform chunk retrieval across given noisy document sets. In this setting, the system loses capabilities as follows: (1) strategic document filtering (critical for M4DocBENCH’s 12.7-document average for each question), and (2) query-adaptive granularity selection. This confirms that multi-granular representations require intelligent selection to be effective.

Effect of Iterative Search. Figure 3 demonstrates how iterative refinement progressively improves retrieval and answer quality. Document-level recall jumps from 62-65% (turn 1) to 75-82% (turn 3), then plateaus at approximately 82-83% by turn 5, indicating that the search-refine loop effectively identifies relevant documents within 2-3 iterations. Accuracy exhibits substantial gains: Qwen3-32B improves by 7.2%, Qwen3-235B by 9.7%, and DeepSeek-R1 by 11.1% from turn 1 to turn 5. However, the steepest improvements occur between turns 1-3, after which gains diminish despite accumulating additional chunks (from 7-10 at turn 1 to 14-17 at turn 5), suggesting increasing redundancy. Notably, stronger models achieve higher accuracy with fewer chunks, indicating more efficient evidence selection. Fine-grained layout recall follows similar patterns, improving from 24-26% to 40-45%. These results validate our iterative search design while suggesting that 3 iterations offer an optimal balance between accuracy and efficiency.

5.6 Parsing and Deep Research Efficiency

Table 4 analyzes computational tradeoffs across parsing strategies on M4DocBENCH’s 304 documents (6,177 pages, 4,146 figures, and 2,739 tables) and the averaged deep research latency measured on total 158 questions in M4DocBENCH.

Offline Doc-Parsing Breakdown. The offline processing stage reveals substantial time-cost trade-offs across parsing strategies. Parsing-free approaches offer the fastest setup, requiring only indexing. Shallow-parsing introduces a modest parsing step (18:47) with similar indexing time (9:50), resulting in a total offline cost of approximately 28 minutes. In contrast, deep-parsing demands substantially greater upfront investment, with parsing (1:21:21) and captioning (1:12:32) combining for 2.5 hours of preprocessing time.

Online Deep-Research Breakdown. For deep research workflows, latency distributes across planning, iterative search-refine loops, and answer generation. The search-refine phase dominates due to multiple retrieval rounds and evidence accumulation, accounting for 80%, 60%, 50% of total latency for parsing-free, shallow-parsing, and deep-parsing respectively.

Parsing-time vs. Research-time Trade-off. Deep parsing requires around 2.5h upfront: much slower than parsing-free (9-39m), which primarily due to MinerU’s layout analysis and visual element captioning. However, this investment dramatically reduces research-time latency. Parsing-free methods spend significant inference time (07:48) extracting information from raw screenshots, while deep parsing completes the same tasks with 80% less time (01:40). This 5× speedup during research validates that one-time captioning is more efficient than repeated visual processing.

Storage and Embedding Considerations. Multi-vector embeddings consume 10-20× more storage and 4x more indexing time than dense variants, reflecting the tradeoff between retrieval quality and resource requirements. Note that multi-vector embedding storage and similarity computation (maxsim in late interaction mechanism) is not supported in current vector database, thus causing much longer indexing and searching latency.

Production-scale deployment. We deploy our vector database at production scale with over 10-million pages in corpus. The retrieval latency (combining query indexing time and matching time using inner-product and IVF_FLAT index) remains 200ms per query. In comparison, querying over 1-million pages cost 150ms.

6 Conclusion

We introduced Doc-Researcher, addressing the critical limitation that existing deep research systems cannot process multimodal documents. Our work identifies and resolves three fundamental gaps in current systems. First, we developed a deep multimodal parsing framework that preserves layout structure and modality-specific characteristics, achieving 11.4% improvement over simplistic OCR approaches. Second, we designed a systematic retrieval architecture supporting multiple paradigms (text/vision/hybrid) and granularities (chunk/page/full/summary). Third, we implement iterative multi-agent workflows that mirror human research processes through planning, search-refine loops, and progressive synthesis, leading to 3.4× better performance than existing baselines. Beyond the system, we contributed M4DocBench, the first benchmark evaluating multimodal, multi-hop, multi-document, and multi-turn research capabilities with 158 expert-annotated questions spanning 4 domains. In the future, we will explore three directions: developing compact multimodal embeddings to reduce computational overhead, improving cross-modal reasoning mechanisms for better evidence synthesis, and extending to active learning that adapts retrieval strategies based on user interaction patterns. These advances would further enhance the efficiency and effectiveness of multimodal document research.

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Metric	Education	Finance	Insurance	Research	Overall
Question Statistics : <i>at least one evidence, evidence of multiple modalities and from multi-documents, multi-turn conversations</i>					
#Question	30	36	24	68	158
TXT/TAB/I	24 / 22 / 0	34 / 24 / 28	18 / 22 / 12	50 / 20 / 30	126 / 88 / 70
Multi-mod/doc/turn	22 / 30 / 4	36 / 36 / 6	18 / 22 / 16	36 / 56 / 32	112 / 144 / 58
Document Statistics per Question : <i>min / max / median / avg</i>					
#Docs (total)	4 / 16 / 8 / <u>7.8</u>	9 / 12 / 9 / <u>10.0</u>	9 / 42 / 11 / <u>13.8</u>	5 / 22 / 17 / <u>15.9</u>	4 / 42 / 11 / <u>12.7</u>
#Docs (relevant)	2 / 6 / 2 / <u>3.0</u>	3 / 7 / 3 / <u>3.9</u>	1 / 10 / 3 / <u>4.6</u>	1 / 10 / 4 / <u>3.8</u>	1 / 10 / 3 / <u>3.8</u>
Evidence Statistics per Question : <i>min / max / median / avg</i>					
#Total Page	8/174/55/ <u>72.5</u>	151/357/207/ <u>222.6</u>	123/461/165/ <u>195.8</u>	132/836/484/ <u>493.0</u>	8/836/228/ <u>306.4</u>
#Relevant Page	2 / 8 / 3 / <u>3.3</u>	3 / 19 / 6 / <u>6.7</u>	2 / 31 / 4 / <u>9.6</u>	1 / 19 / 7 / <u>7.9</u>	1 / 31 / 5 / <u>7.0</u>
#Layout	2 / 8 / 5 / <u>4.9</u>	6 / 48 / 12 / <u>14.8</u>	2 / 40 / 6 / <u>11.7</u>	1 / 157 / 16 / <u>20.6</u>	1 / 157 / 8 / <u>14.8</u>
Evidence Modality Distribution : <i>text (body text, title, equations, etc) / table / image (figure, chart, infographic, etc)</i>					
% by #Chunk	52.4 / 47.6 / 0.0	44.0 / 38.0 / 18.0	49.6 / 42.6 / 7.8	83.8 / 6.8 / 9.4	67.5 / 16.6 / 15.9
% by Bbox Area	28.2 / 71.8 / 0.0	41.5 / 22.1 / 36.4	9.2 / 83.8 / 7.0	69.0 / 18.4 / 12.6	40.9 / 30.2 / 28.9
Answer Statistics: length (words) : <i>min / max / median / avg</i>					
Short Answer	13 / 303 / 26 / <u>49.1</u>	35 / 383 / 85 / <u>108.7</u>	13 / 80 / 50 / <u>48.4</u>	5 / 117 / 32 / <u>36.1</u>	5 / 383 / 44 / <u>59.7</u>
Long answer	40/747/122/ <u>174.7</u>	185/698/478/ <u>464.8</u>	115/650/266/ <u>314.2</u>	23/550/112/ <u>183.4</u>	23/747/223/ <u>272.0</u>
Other Statistics : <i>min / max / median / avg</i>					
#Multi-turn	2 / 2 / 2 / <u>2.0</u>	2 / 2 / 2 / <u>2.0</u>	2 / 4 / 3 / <u>2.6</u>	2 / 3 / 2 / <u>2.2</u>	2 / 4 / 2 / <u>2.3</u>
#Search clauses	2 / 8 / 3 / <u>3.7</u>	2 / 12 / 5 / <u>5.2</u>	3 / 10 / 5 / <u>4.8</u>	1 / 6 / 3 / <u>3.1</u>	1 / 12 / 3 / <u>3.9</u>

Table 5: Statistical details of M4DocBENCH by domains.

A M4DocBENCH: Supplementary Materials

Statistical Details. In this section, we provide more fine-grained statistics of M4DocBENCH, which supplements analysis in Section 4.4. In Table 6, we report the statistical details about multi-modal documents in M4DocBENCH. Note that documents of all domains contain considerable figures and tables. In Table 5, we provide more fine-grained statistics of M4DocBENCH by domains as well as statistics of question, document, evidence, answer, etc.

Annotation Demonstration. Although Figure 1 shows a typical annotation of M4DocBENCH, we provide two more annotation examples as shown in Figure 4. This is to facilitate readers' understanding on M4DocBENCH and its annotation process in Section 4.3.

Domain	Doc Count	Avg Page	Avg Word	Avg Fig.	Avg Tab.	Avg Eq.
Research	166	26.7	21.8k	17.3	11.5	2.1
Insurance	43	11.0	5.5k	10.8	5.4	0.0
Education	56	7.5	3.9k	4.3	4.3	0.0
Finance	38	22.2	11.2k	14.4	9.5	0.0
Overall	303	20.4	14.8k	13.6	9.0	1.2

Table 6: Document statistics across different domains

B Implementation Details

Multimodal Document Parsing. We utilize MinerU v2.0.6 for document parsing. The chunking strategy segments text when: (1) length exceeds 300 tokens, (2) page boundaries are crossed, or (3) section headers are encountered. Images and tables are extracted as independent units. **Image Processing:** Visual content was processed using Qwen2.5-VL-7B-Instruct² for description generation. Visual chunks of image size less than 10KB are excluded. This is to filter decorative icons and insignificant graphics, as well as relieving computational overhead. **OCR Processing:** PDF pages were

²Qwen2.5-VL-7B-Instruct is served using vLLM for efficient and stable inference.

	Model	Size	Base Model	HuggingFace Checkpoint
Text	BM25	-	-	-
	BGE-M3	0.56B	XLm-RoBERTa-large	BAAI/bge-m3
	E5	0.34B	BERT-large	intfloat/e5-large-v2
	Qwen3-embed	0.6B	Qwen3-0.6B-Base	Qwen/Qwen3-Embedding-0.6B
Visual	Qwen3-reranker	0.6B	Qwen3-0.6B-Base	Qwen/Qwen3-Reranker-0.6B
	DSEwiki-ss	4.15B	Phi-3-Vision	Tevatron/dse-phi3-v1.0
	ColPali	2.92B	PaliGemma	vidore/colpali
	ColQwen2	2.21B	Qwen2-VL-2B-Instruct	vidore/colqwen2-v1.0
	Jina-embed-v4	3.75B	Qwen2.5-VL-3B-Instruct	jinaai/jina-embeddings-v4

Table 7: Checkpoints for Text and Vision Retrievers.

converted using PyMuPDF (fitz) and subjected to Tesseract OCR with a chunk size of 140 characters for structured text extraction.

Multimodal Indexing and Data Persistence. We deploy MySQL database to store document metadata, including document summaries, structured full-text, chunks (both text and cropped images), pages (both text and screenshots), and layout bounding boxes (bbox coordinates). We summarize the checkpoints for retriever deployments in Table 7. Retrievers are used to convert passages of different modalities and granularities into vector representations. We deploy Milvus vector database to maintain dense vectors for textual and visual chunks, and page screenshots. Document names serve as filtering conditions for retrieval operations. For *multi-vector storage*, we save them locally as np.savez_compressed, with each document stored as an individual NPZ file for efficient retrieval.

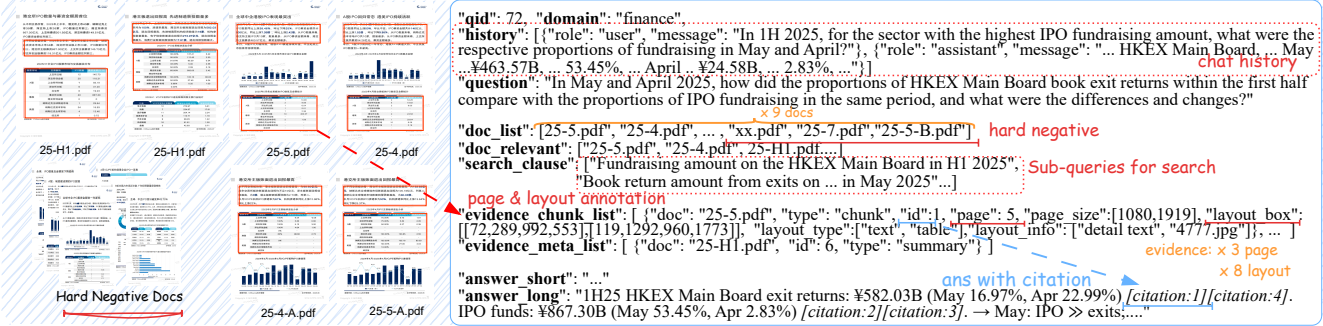
Deep Research Framework. We implement DOC-RESEARCHER, the multi-agent deep research workflow using LangGraph³.

C Extended Experimental Results

Figure 6a depicts the retrieval performance on M4DocBENCH by 4 different domains and 3 different top-*k* settings. Figure 6b depicts the retrieval performance on MMDocIR by 10 different domains and 3 different top-*k* settings. This is extended retrieval results of Table 2.

³<https://www.langchain.com/langgraph>

(a) The question requires integrating textual and tabular evidence from multiple market analysis reports to compare fund-raising and exit-return proportions across different months. The reasoning involves multi-hop retrieval and numerical computation across documents.



(b) This example illustrates a cross-document, cross-modal, multi-hop reasoning task. The model needs to extract and compare structured program information across multiple brochures to determine the shortest graduation time under the given limitation.

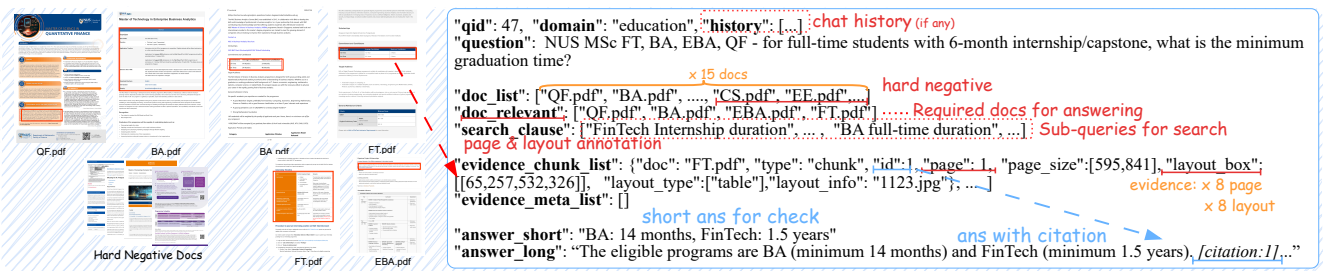


Figure 4: Demonstration of two typical annotations of M4DocBENCH. Note that they are simplified for brevity.

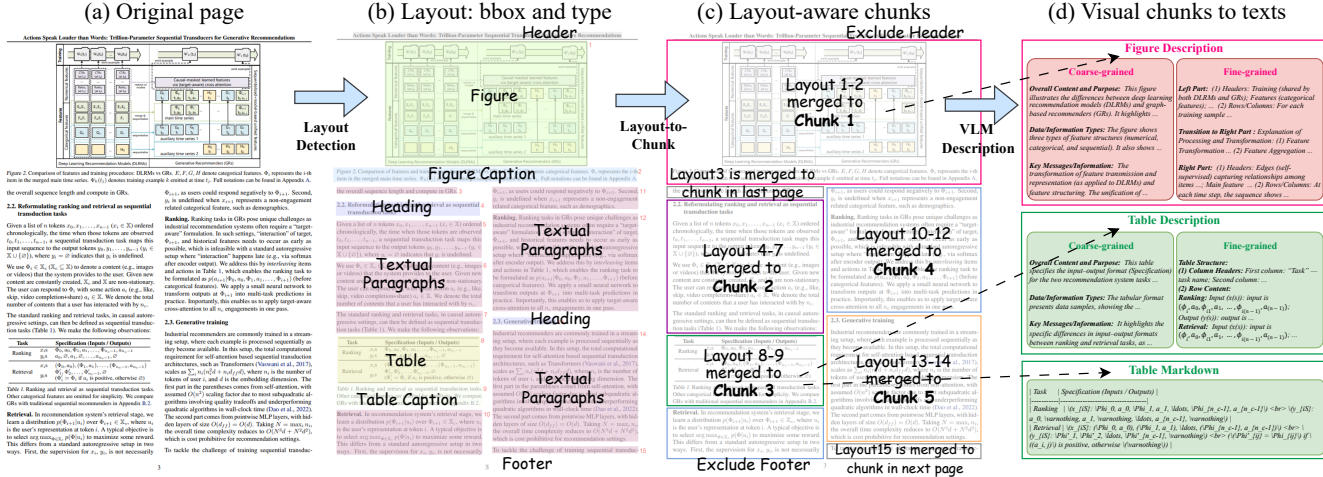


Figure 5: The overview of layout-aware chunking process and multimodal-to-text transcription.

